

Birla Institute of Technology and Science, Pilani
ECE/EEE F311 Communication Systems
First Semester 2025-2026 (01-12-2025)
Comprehensive Exam (Open Book)

Maximum Marks: 120

Duration: 180 minutes

- There are five questions. All questions are compulsory.
- Put important steps and final answer in a rectangular box.
- Answer all the parts of a question at the same place.

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- Q1. (a) Using an adder, a nonlinear (square-law) device, and a band-pass filter centered at f_c , design a block diagram that generates an AM signal for the message signal $m(t)$ of bandwidth B at the carrier frequency f_c with modulation index 1. Assume the message signal has peak amplitude ± 1 and $f_c \gg B$. [8 Marks]
- (b) A message signal $m(t)$ of bandwidth B modulates a unit-amplitude carrier of frequency f_c to produce an upper-sideband suppressed-carrier (USSB) signal $x(t)$. Assume $f_c \gg B$. At the receiver, you have access to $x(t)$, Hilbert filters, and local oscillators that can generate sinusoidal signals at frequency f_c . Using these resources, demodulate the message signal $m(t)$ with a receiver block diagram that implements the required demodulation. Your method must use both $x(t)$ and $x_h(t)$ (the Hilbert transform of $x(t)$). [8 Marks]
- (c) A wide-sense stationary (WSS) real-valued baseband signal $x(t)$ has an autocorrelation $R_x(\tau) = 10 \text{sinc}(10\pi\tau)$. The signal $x(t)$ is received in additive white Gaussian noise $n(t)$ whose double-sided power spectral density is $N_0/2 = -60 \text{ dBm/Hz}$. Determine the signal-to-noise ratio (SNR) in dB for this system. [8 Marks]
- Q2. You have to design an analog communication system for your campus. A 4 MHz channel centered at 50 MHz (48–52 MHz) is available for radio broadcasting. The channel introduces zero delay but has a signal amplitude attenuation of 10^{-1} (i.e., a transmitted amplitude of 10 becomes a received amplitude of 1). Assume additive noise to be zero. Solve the following two design problems:
- (a) You have to design two AM transmitters: transmitter 1 (near the Clock tower) for $m_1(t) = 20 \sin(2\pi 10^6 t)$ and transmitter 2 (near the main entrance gate) for $m_2(t)$, where $m_2(t)$ has a bandwidth of 1 MHz and has the same power as $m_1(t)$. Assume zero guard band between two stations. You are also required to design a single receiver such that the receiver can select a desired station (here transmitter 1) to demodulate $m_1(t)$. The receiver must be inexpensive. At the receiver, you must use at least two band-pass filters. For efficient demodulation, the received signal power at the front end of the receiver (i.e., after channel output) should be 10 W.

Considering the above parameters, design a communication system for two stations transmitting $m_1(t)$ and $m_2(t)$ by specifying the following:

- (i) Transmission equations for both stations
- (ii) Transmit power
- (iii) Transmitter block diagrams
- (iv) Receiver block diagram with numerical parameter of each block to demodulate the signal $m_1(t)$.

[12 Marks]

- (b) You have to design a single FM transmitter: transmitter 1 (near the Clock tower) for an another message signal $m(t) = \cos(2\pi 10^6 t)$. You are given a FM transmitter FM0 with a maximum frequency deviation of 10 KHz. The FM0 cannot be directly used. Using FM0, design a single FM transmitter FM1 such that FM1 occupies the full 4 MHz channel bandwidth (48–52 MHz) for the message $m(t)$.

You may use any required components or signal-processing blocks for your design.

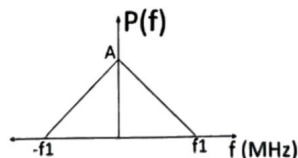
Construct an FM transmitter and specify the following:

- (i) Transmission equation for FM1
- (ii) Complete FM transmitter block diagram
- (iii) Transmitted power

[12 Marks]

Q3. A total of 1.4 billion RAW data bits (1.4×10^9) must be transmitted securely over a band-limited channel whose available spectrum is restricted to 3.2 MHz. Because of this bandwidth constraint, multiple bits will necessarily overlap in both time and frequency, which can give rise to ISI. To avoid the harmful effects of ISI while still allowing all 1.4 billion bits to overlap in the same time–frequency resource, the RAW data must be transmitted using an appropriate pulse-shaping strategy. Solve the following:

- (a) For binary polar transmission, consider the bit stream 1010. Using an optimal ISI-free pulse, draw the time-domain waveform and the corresponding magnitude spectrum for the 4-bit sequence, clearly marking the time and frequency axes with appropriate numerical labels for all four transmitted bits. Also write down the explicit time-domain expression of the pulse $p(t)$ corresponding to a single bit, and its frequency-domain representation $P(f)$. [8 Marks]
- (b) Instead of using an optimal pulse, consider a pulse whose frequency spectrum $P(f)$ is shown in the figure below. Determine the numerical values of $-f_1$, f_1 , and the amplitude level A required to satisfy the zero-ISI criterion. Furthermore, assuming a noise-free channel, compute the achievable data rate (in bps) when 64-ary QAM is used. [8 Marks]

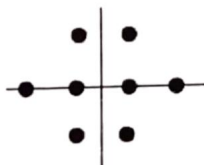


- (c) Assuming a signal power of 10 W, and double-sided noise PSD $N_0/2 = 10^{-8}$ W/Hz, determine an appropriate M -ary PSK constellation size M that can operate reliably within the given 3.2 MHz bandwidth over an AWGN channel. Using this value of M , and assuming the optimal pulse shape is used, compute the minimum time required to transmit all 1.4×10^9 bits. How much time it will take if $B \rightarrow \infty$? [8 Marks]

- Q4. (a) A polar encoded bit stream is transmitted using rectangular pulses of width 1 second and amplitude 5, where a transmitted bit 1 corresponds to the pulse $p(t)$ and a bit 0 corresponds to $-p(t)$, with equal probabilities. The channel has impulse response $\delta(t)$, and the additive noise is AWGN with double-sided PSD $N_0/2 = 10$ W/Hz. The receiver front-end bandwidth is taken as the main-lobe bandwidth of the rectangular pulse and a matched filter is employed.

Draw the impulse response of the matched filter and sketch the corresponding matched-filter outputs for the transmission of a polar bit 1 and a polar bit 0. Determine the SNR at the input of the matched filter and the SNR at the output of the matched filter. [12 Marks]

- (b) An 8-QAM constellation with equiprobable symbol transmission is shown in the figure below. Using the basis-signal representation, draw the complete transmitter and receiver block diagram employing matched filter over an AWGN channel with noise PSD $N_0/2$. Clearly indicate the input and output signals at each stage of both the transmitter and the receiver by taking an example of a constellation symbol. Find the probability of correctness of a symbol with the least energy in terms of E_b/N_0 . Note: Ignore the effect of diagonal noise if any. [12 Marks]



- Q5. (a) A discrete memoryless source emits 5 symbols with unequal probabilities and is encoded using a binary prefix code. The first four codewords have lengths 2, 3, 4, and 4 bits. The code is designed to be optimal in the sense that it achieves the minimum possible average codeword length for the given symbol probabilities. Determine the length of the fifth codeword. [6 Marks]
- (b) A discrete memoryless source emits two symbols m_1 and m_2 with probabilities $P(m_1) = 0.9$ and $P(m_2) = 0.1$. Construct the Shannon-Fano codes for the second-order and third-order source extensions, and compare the coding efficiencies obtained in both cases. Verify that each constructed code satisfies the prefix condition. Finally, determine the efficiency of an n -th order Shannon-Fano code as $n \rightarrow \infty$. [6 Marks]

- (c) Consider the following three discrete memoryless channels with the given transition probability matrices:

$$C_1: P_1(Y|X) = \begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.05 & 0.9 & 0.05 \\ 0.05 & 0.05 & 0.9 \end{bmatrix}, \quad C_2: P_2(Y|X) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$C_3: P_3(Y|X) = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.8 & 0.1 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}.$$

Assume that, in each case, the input symbols are transmitted with equal probability. Arrange the three channels C_1, C_2 , and C_3 in order of increasing channel capacity with justification. Also, determine the maximum channel capacity among these three channels in bits per symbol. [6 Marks]

- (d) A (7, 3) binary linear block code is defined by the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}.$$

List all the codewords of the code and determine its error-detection capability.

A received vector

$$r = 1001010,$$

is known to contain exactly one bit error. Using Hamming-distance decoding (as explained in the following note), determine the most likely transmitted codeword.

Note: The Hamming weight $w(s)$ of a binary vector s is the number of 1s in s . The Hamming distance between two codewords s_i and s_j is

$$d(s_i, s_j) = w(s_i \oplus s_j),$$

where $\oplus$ denotes modulo-2 addition. Thus, compute the Hamming distance between the received vector and each codeword; the codeword with the minimum Hamming distance is taken as the decoded one. [6 Marks]
